General Sir John Kotelawala Defence University



Faculty of Computing

Essentials of Artificial Intelligence

Final Report

AI-Based Dengue Breeding Site Detection and Risk Prioritization System

Group 58	
Team Members	**Index Number**
D/BIT/24/0043	WWSN Wickramarachchi
D/BSE/25/0025	KGT Nethmina
D/BIT/24/0074	H.K.D.C Akalanka
D/COE/25/0003	W.M.D.G.Janani Warnasooriya

Table of Contents

Abstract **3**
1. Introduction **4**
2. Problem Background **5**
4. Objectives **7**
4.1 Main Objective 7
4.2 Specific Objectives 7
5. Selected AI Techniques **8**
5.1 Computer Vision 8
5.2 YOLO-Based Object Detection 8
5.3 Risk Scoring 9
6. Justification of AI Techniques **11**
6.1 Why Computer Vision? 11
6.2 Why YOLO? 11
6.3 Why Risk Scoring? 11
6.4 Why Human Verification? 11
7. System Design **12**
7.1 Overall System Architecture 12
7.2 System Workflow 13
8. Dataset **14**
8.1 Dataset Preparation 16
9. AI Model Structure **17**
10. Implementation Details **18**
10.1 Frontend 18
Figure 13: User image upload interface 18
10.2 Backend 18
10.3 Database 21
10.4 GPS and Location Handling 22
Figure 19: GPS Location/ Map Interface 22
10.5 Weather and Historical Data Integration 22
11. Testing and Results **23**
11.1 Functional Testing 23
11.2 AI Detection Testing 24
11.3 Risk Prioritization Results 25
12. Model Evaluation **26**
Suggested table: 27
13. Limitations and Discussion **28**
13.1 Dataset Size 28
13.2 Image Variations 28
13.3 Stagnant Water Detection 28
13.4 False Positives and False Negatives 28

13.5 External Data Availability 28
14. Ethical and Social Considerations 29
14.1 Privacy 29
14.2 Data Security 29
14.3 AI Bias 29
14.4 Human Oversight 29
14.5 Responsible Use of AI 29
15. Conclusion 30
16. References 31
17. Individual Contribution 32

Abstract

Dengue is still a public health issue in many places. Stopping mosquitoes from breeding matters a lot for keeping it under control. The hard part seems to be spotting the right locations and deciding which ones to check first when reports start piling up. Manual inspections just do not scale well then.

This project outlines an AI system meant to help public health inspectors with that. It takes images from users and runs them through computer vision tools based on YOLO to find things like standing water or containers that could collect it. Tires and buckets come up often in the description along with pots and bottles.

The detections get paired with location data and weather details such as rainfall plus past dengue records to assign some kind of risk level. Reports then show up on a dashboard for the inspectors to review and verify in person. It is presented as a support tool rather than anything that diagnoses the disease itself so the human check stays essential at the end.

The prototype shows how image analysis and geographic information can work together for prevention work. I am not totally sure how every data source gets weighted in the scoring but the overall flow makes sense for reducing the manual load. Some of the object types listed feel like they could overlap in practice.

Keywords: Artificial Intelligence, Dengue, Mosquito Breeding Sites, Computer Vision, YOLO, Object Detection, Risk Scoring, GPS, Public Health

1. Introduction

Dengue is a disease caused by a virus carried through mosquitos which poses a great threat to public health in many tropical and sub-tropical countries. The prevention of this disease involves reduction of environments that can harbor mosquitos, especially locations with standing water.

Identification of breeding sites has traditionally involved physical inspection and reporting of breeding sites by the people. If there are too many reports, then it becomes challenging for Public Health Inspectors to find out the sites that need urgent action.

The proposed project solves this problem through creating an AI-supported system that will analyse images reported by the users. The system will be able to recognize visual features of potential breeding sites and integrate AI results with geographical and environmental data.

The main focus of the project is the identification of potential breeding sites and risk prioritization rather than detecting infection and diagnosing the disease.

In the final system, the following elements are combined:

- Computer vision
- Object detection (based on YOLO)
- Image processing
- Location/GPS information
- Weather information
- Dengue historical information
- Risk scoring
- Priority classification
- PHI verification

The use of AI helps public health officials in their work while maintaining the role of human verification in the process.This follows the project's original refinement from general “dengue detection” toward potential breeding-site detection and risk prioritization.

2. Problem Background

For dengue prevention, it is essential to detect and get rid of potential mosquito breeding sites.

These could include discarded tyres, buckets, bottles, flower pots, coconut shells, and any other items that are capable of holding water.

While in real life, these potential locations could be spread throughout a vast geographical space, PHIs would likely receive several complaints or reports about these, however, not all locations will carry the same risk.

If there was a system capable of automatically analysing images sent to it and providing preliminary risk assessment, it would help PHIs concentrate their time and effort on more urgent cases.

That is why this project combines the use of image recognition AI along with some context.

According to the progress report, accumulated water, water-filled tyres, containers, buckets, flower pots and other water-containing objects were listed among the visual cues.

Figure 1: Examples of potential mosquito breeding-site environments.



3. Problem Statement

Now, locating any areas that could be used for breeding by the mosquitoes could involve manual inspection and reports from people. In cases where there are many reports, PHIs could face challenges as to which areas should be inspected first.

Moreover, the pictures taken by the users could have other objects and other lighting, camera angles, background, and the water present within the pictures could also be different. Manually going through pictures could prove to be a time-consuming process.

This challenge could be solved through the use of the AI-based system which would analyze the pictures and look for the signs of any potential breeding spots. These pictures would be analyzed along with location, weather, rainfall, and the information about dengue to generate the risk scores and priorities.

This system will not detect dengue or even mosquitoes. It will just provide decision-support information to PHIs.

4. Objectives

4.1 Main Objective

Design an AI-powered system that will analyze submitted photos to detect visual clues indicating the possible breeding grounds of mosquitoes and sort out the reported areas based on their risk rating.

4.2 Specific Objectives

Specific objectives will be as follows:

1. To gather and prepare a dataset with images of the possible mosquito breeding locations.
2. To tag all objects and visual cues found in the gathered images.
3. To create a YOLO based object detection algorithm capable of detecting the potential breeding site indicators.
4. To create a system for uploading the images taken by the user.
5. To harvest geographical data connected to the submitted report.
6. To incorporate the weather and historical information about dengue.
7. To develop a scoring system taking into account multiple variables.
8. To sort the submissions into low, medium, and high risk categories.
9. To create a dashboard for the review and prioritizing of the submissions made by the user.
10. To retain human approval as the final decision-making step.

5. Selected AI Techniques

The project uses several AI and data-processing techniques.

5.1 Computer Vision

Computer Vision is used to analyze the images that have been uploaded by the user.

These include trying to detect visual features such as:

- Tyre
- Coconut-Exocarp
- Water-filled Container
- Flower Pot
- Drain-Inlet

Other containers for water

The objective is not to detect whether water has definitely stagnated for a certain amount of time. The objective is to detect features that could act as a breeding environment.

5.2 YOLO-Based Object Detection

A YOLO-based object detection algorithm is employed for the AI detection process.

The algorithm takes in an image input and detects the relevant objects through the use of bounding boxes and confidence levels.

For example:

Input:
Photograph of a discarded tyre containing water.

Possible AI output:

- Tyre → 96%
- Accumulated Water → 91%

The model output can therefore provide:

- Object class
- Bounding-box location
    - Confidence score

The project's progress report selected YOLO because the system requires object detection rather than only image-level classification.

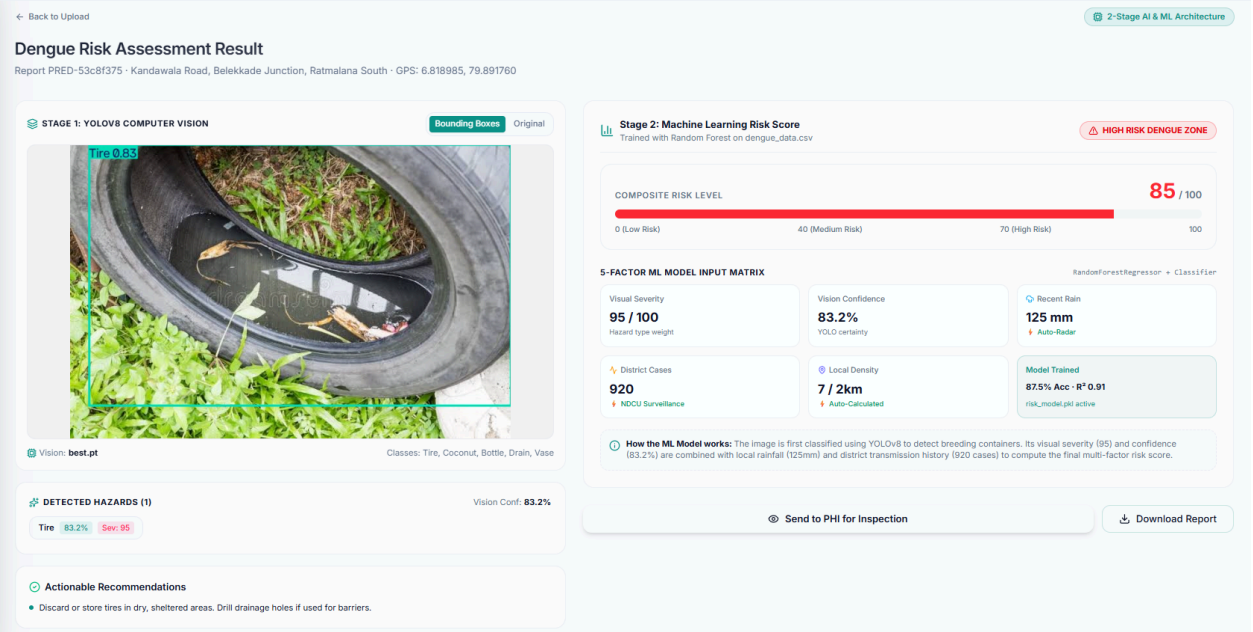


Figure 2: YOLO-based detection of potential breeding-site indicators.

5.3 Risk Scoring

The results of the AI detection are supplemented with further data for computing a total risk score.

The data used are:

- AI detection result
- Location
- Weather conditions
- Rainfall
- Dengue history

Based on the score, the reports are divided into five priority categories.

- Critical (81-100)
- High (61-80)
- Moderate (41-60)
- Low (21-40)
- Minimal (0-20)

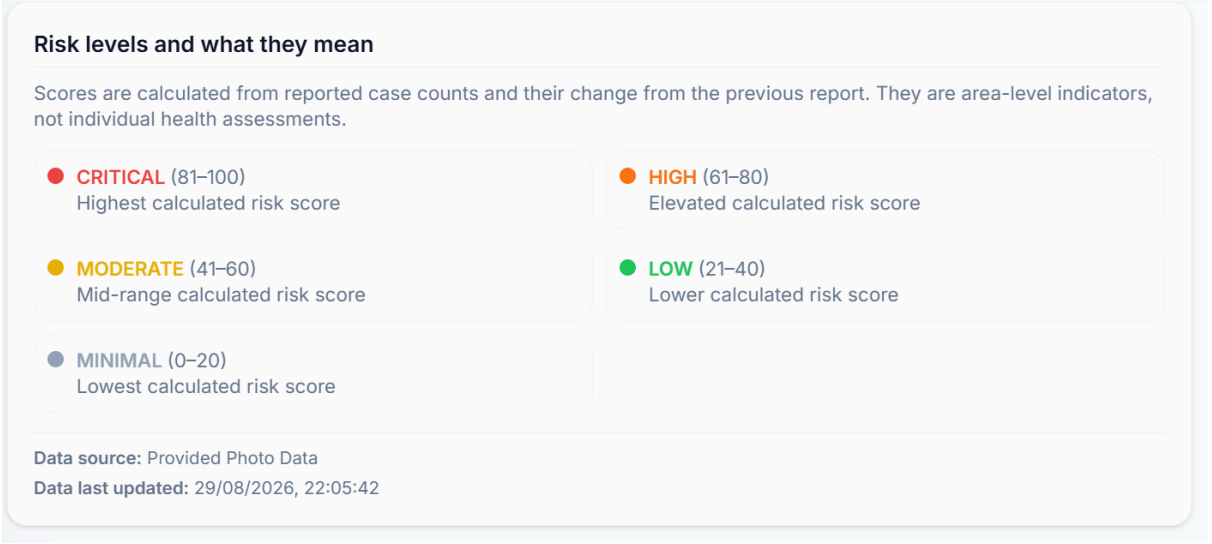


Figure 3: Rick Scoring

The exact contribution of individual factors may be adjusted based on system testing and evaluation.

AI Detection + GPS + Weather + Rainfall + Historical Data → Risk Score → Risk Level → PHI Priority

6. Justification of AI Techniques

6.1 Why Computer Vision?

The input into the system is an image. Hence, Computer Vision is suitable in capturing the visual information contained in the image.

A possible breeding environment could have some visual cues, such as tyres, drums, bottles, and stagnant water.

6.2 Why YOLO?

YOLO is appropriate since the project involves object identification from an image.

While YOLO does not predict the presence of an object in an image, it can help find out the location of the object in an image.

This is helpful especially if there are many objects present in an image.

In an example, an image may have:

- Tyre
- Plastic bottle
- Bucket
- Water pooled up

6.3 Why Risk Scoring?

Detection of AI on its own cannot offer enough information on how to prioritize one particular report over another.

Two reports can have the same objects, but they can be located in places that have different environmental and historical background.

This is where the risk scoring helps in determining the correct priority for the PHIs.

6.4 Why Human Verification?

AI systems may generate false positives and false negatives. Consequently, an AI-generated prediction cannot be assumed to be an absolute positive breeding site.

AI is employed in the system as a decision-making tool, with the PHIs being responsible for verifying the site physically.

7. System Design

7.1 Overall System Architecture

The system consists of the following major components:

1. User Interface
2. Backend/API
3. Image Processing
4. AI Object Detection Model
5. Risk Scoring Component
6. Weather and Historical Data
7. Database
8. PHI Verification Interface

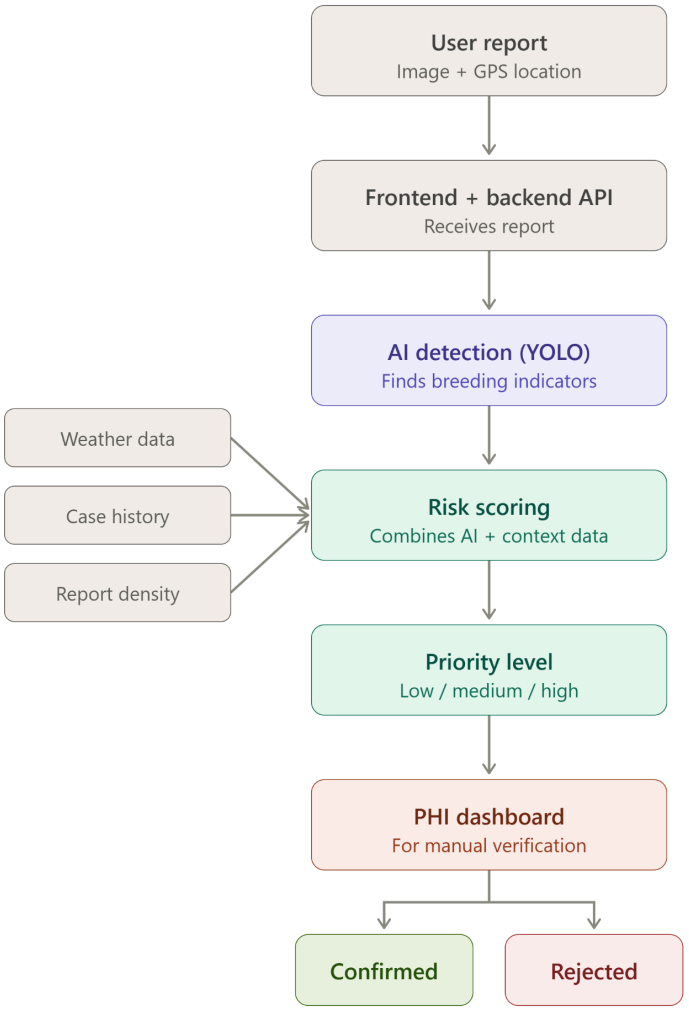


Figure 4: Complete System Architecture

The overall architecture follows the workflow from user submission through AI analysis and risk prioritization to PHI verification.

7.2 System Workflow

The main system workflow is:

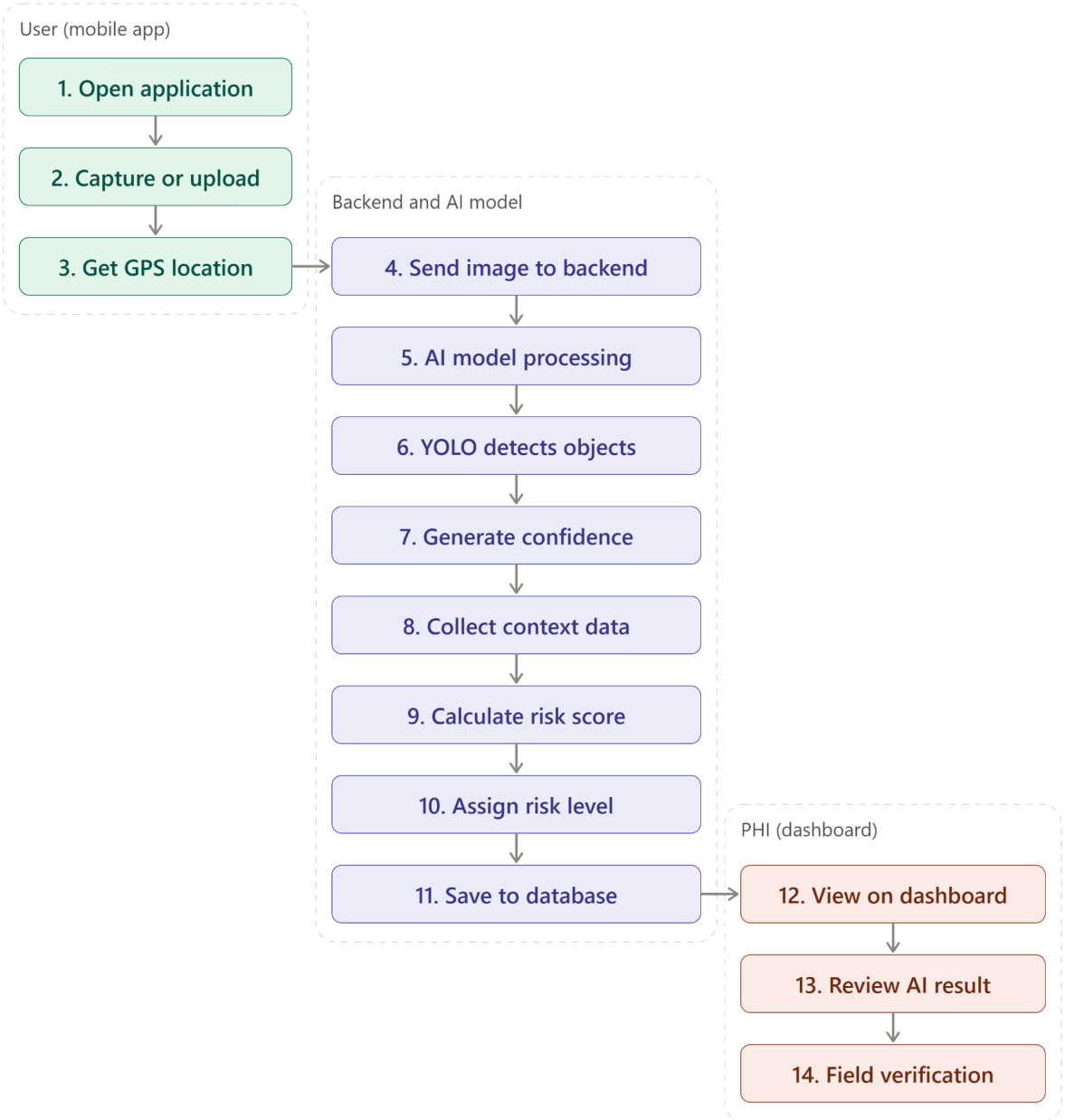


Figure 5: System Workflow

8. Dataset

A data set was created to train and test the AI algorithm. The data set is based on images that depict both breeding grounds of mosquitoes and non-breeding grounds of mosquitoes.

The dataset categories include:

Category	Example
Tyre	Discarded tyres in outdoor environments
Coconut-Exocarp	Discarded king coconut or regular coconut shells left outdoors
Water-filled Container	Buckets, bottles and containers containing water
Flower Pot	Flower pots containing accumulated water
Drain-Inlet	Blocked drain inlets and gutters collecting leaf debris and water



Figure 6: Tire



Figure 7: Coconut-Exocarp



Figure 8: Bottle



Figure 9: Drain-Inlet



Figure 10: Vase

8.1 Dataset Preparation

The dataset preparation process consisted of:

1. Image collection
2. Image selection
3. Image cleaning
4. Image labelling
5. Dataset organization
6. Training/validation/testing split
7. Model training
8. Model evaluation



Figure 11: Dataset labelling showing bounding boxes for the 'Coconut-Exocarp' class



Figure 12: Dataset labelling showing bounding boxes for the 'Vase' class

9. AI Model Structure

The AI component uses a YOLO-based object detection architecture.

The general process is:

Input Image → Image Processing → YOLO Model → Object Detection → Confidence Score → Potential Breeding-Site Indicator

For example:

Input:
Photograph of discarded tyre.

Output:

Tyre → 96%

Accumulated Water → 91%

Decision:

Potential Breeding Site → YES

The confidence score represents the model's confidence in its object detection and is **not** a measurement of dengue infection or medical risk.

10. Implementation Details

10.1 Frontend

The frontend provides the interface through which users can submit reports and PHIs can review submitted reports.

The main interfaces include:

- Image upload
- Report submission
- Location information
- Detection result display
- Risk information
- PHI dashboard

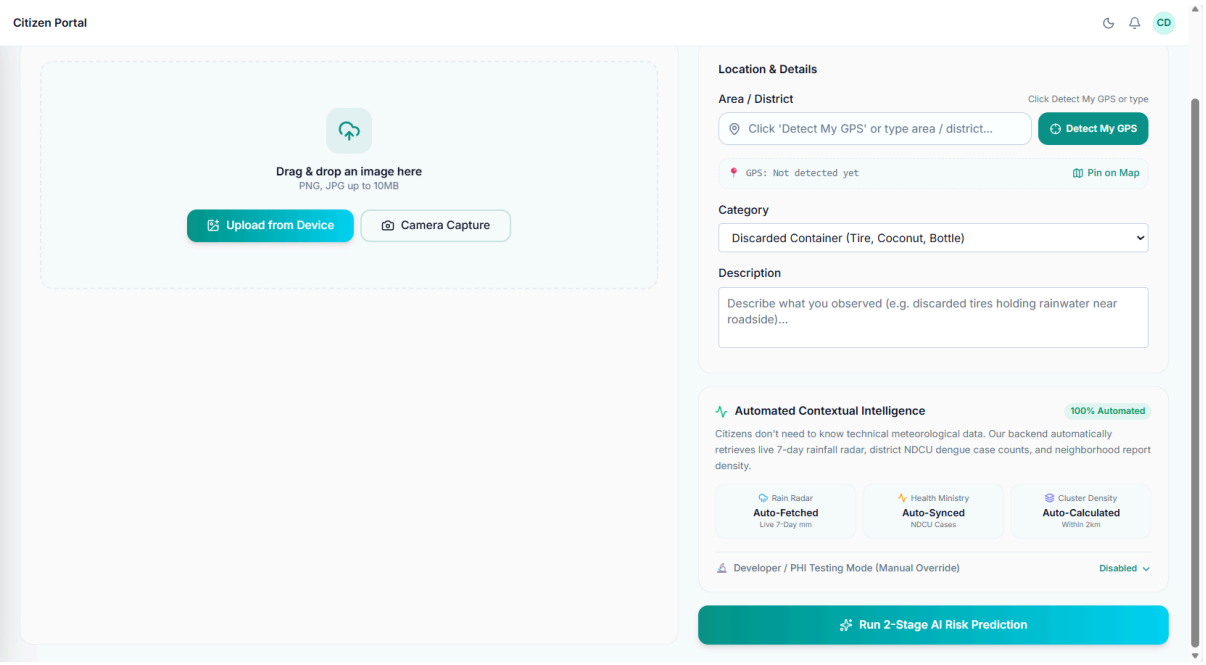


Figure 13: User image upload interface

10.2 Backend

The backend manages communication between the frontend, AI model, database, and external information sources.

The backend is responsible for:

- Receiving user reports
- Processing submitted images
- Communicating with the AI model
- Managing report information
- Storing results
- Providing information to the PHI dashboard

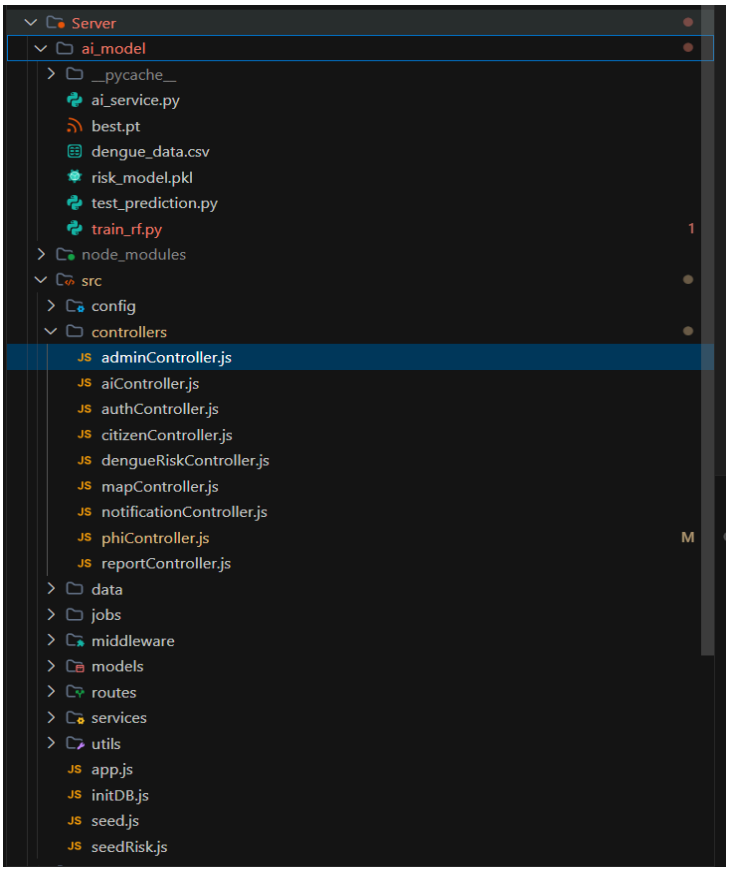


Figure 14: Backend directory structure showing the integration of Node.js controllers and the Python AI microservice

```python
@app.get("/health")
def health():
    bundle = get_rf_model()
    return {
        "status": "online",
        "service": "DengueGuard AI Model",
        "yolo_classes": model.names,
        "rf_model_loaded": bundle is not None,
        "rf_classes": bundle["classes"] if bundle else [],
    }
```

Figure 15: The /health GET endpoint verifying the status of YOLO and Random Forest models

```python
@app.post("/retrain")
def retrain():
    """
    Retrains the Random Forest models on dengue_data.csv and reloads the active model.
    """
    try:
        from train_rf import train_models
        train_models()
        bundle = get_rf_model()
        return {
            "status": "success",
            "message": "Models successfully retrained and reloaded from dengue_data.csv",
            "classes": bundle["classes"] if bundle else []
        }
    except Exception as e:
        raise HTTPException(status_code=500, detail=f"Retraining failed: {str(e)}")
```

Figure 16:The /retrain POST endpoint for dynamically updating the Random Forest risk model

```python
@app.post("/predict")
async def predict(
    file: Optional[UploadFile] = File(None),
    image_path: Optional[str] = Form(None),
    rainfall_mm: float = Form(45.0),
    ndcu_cases: int = Form(320),
    report_density: int = Form(3),
):
    """
```

Figure 17: The core /predict POST endpoint accepting image data and environmental factors for AI inference

10.3 Database

The database stores relevant information related to submitted reports and AI analysis.

Possible stored information includes:

- Report ID
- User information
- Image information
- GPS location
- AI detection result
- Confidence score
- Risk score
- Risk level
- Report status
- PHI verification status

Collection name	Properties	Storage size	Data size	Documents	Avg. document size	Indexes	Total index size
areas	-	36.86 kB	883.00 B	5	176.00 B	5	184.32 kB
denguereports	-	36.86 kB	271.00 B	1	271.00 B	2	73.73 kB
dengueriskS	-	36.86 kB	11.73 kB	27	434.00 B	2	73.73 kB
notifications	-	36.86 kB	5.45 kB	22	247.00 B	6	221.18 kB
phis	-	36.86 kB	676.00 B	1	676.00 B	8	262.14 kB
predictions	-	4.10 kB	0 B	0	0 B	5	20.48 kB
reports	-	36.86 kB	6.30 kB	9	699.00 B	7	258.05 kB
settings	-	36.86 kB	217.00 B	1	217.00 B	2	73.73 kB
users	-	36.86 kB	2.86 kB	7	408.00 B	5	184.32 kB
visits	-	36.86 kB	319.00 B	1	319.00 B	6	221.18 kB

Figure 18: MongoDB database collections and document structure showing stored application data

10.4 GPS and Location Handling

Location information is used to associate each report with a geographical area.

The location can be used together with weather and historical information to improve risk prioritization.

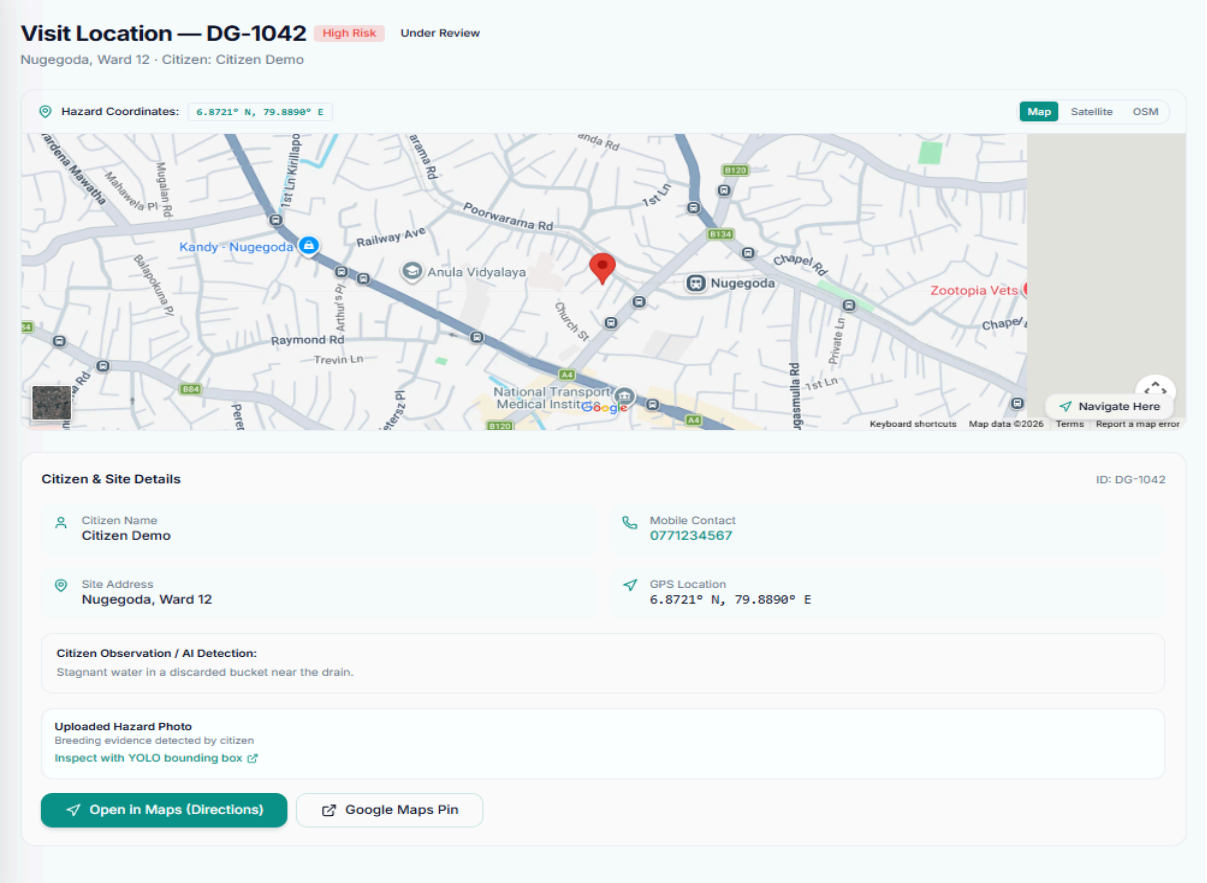


Figure 19: GPS Location/ Map Interface

10.5 Weather and Historical Data Integration

Environmental and historical information can provide additional context for the risk calculation.

The system considers information such as:

- Weather conditions
- Rainfall
- Historical dengue information
- Geographic location

These factors are combined with the AI result rather than relying on the image alone.

11. Testing and Results

Testing was conducted to determine whether the main components of the system function as expected.

11.1 Functional Testing

The following functions were tested:

Test	Expected Result	Result
Upload image	Image successfully submitted	Pass
AI image processing	Image processed by AI model	Pass
Object detection	Relevant objects detected	Pass
Confidence score	Detection confidence displayed	Pass
GPS handling	Report associated with location	Pass
Risk calculation	Risk score generated	Pass
Risk classification	Low/Medium/High assigned	Pass
Report storage	Report stored successfully	Pass
PHI dashboard	Report visible to PHI	Pass

11.2 AI Detection Testing

The model was tested using sample images representing different potential breeding environments.

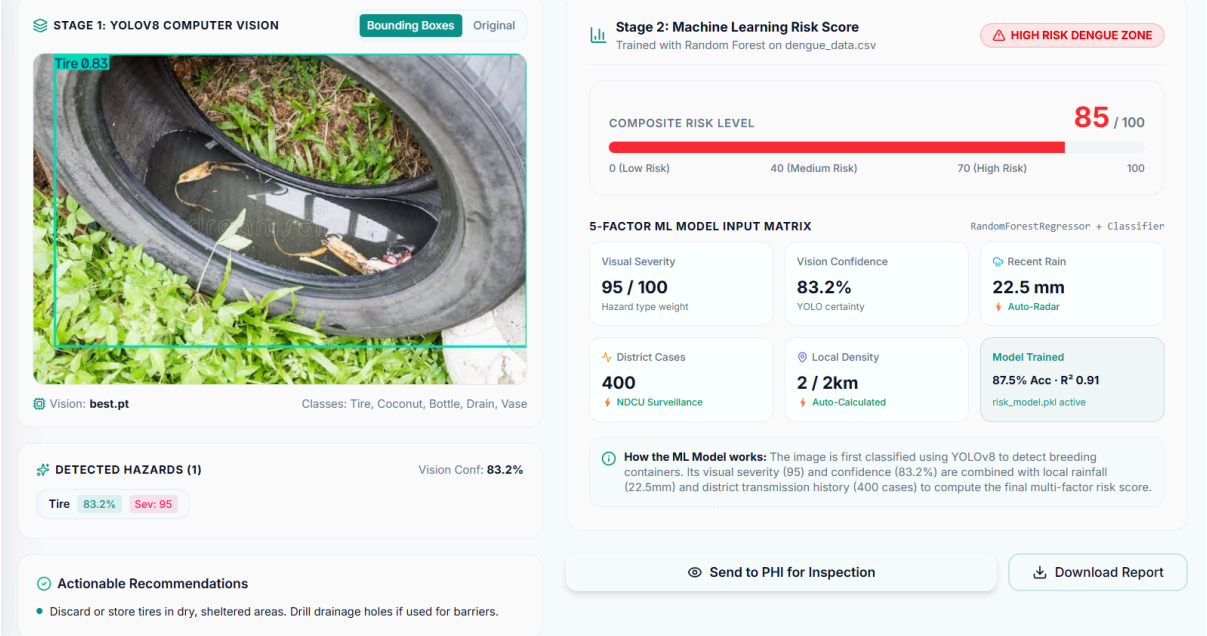


Figure 20: Water-Filled Tyre

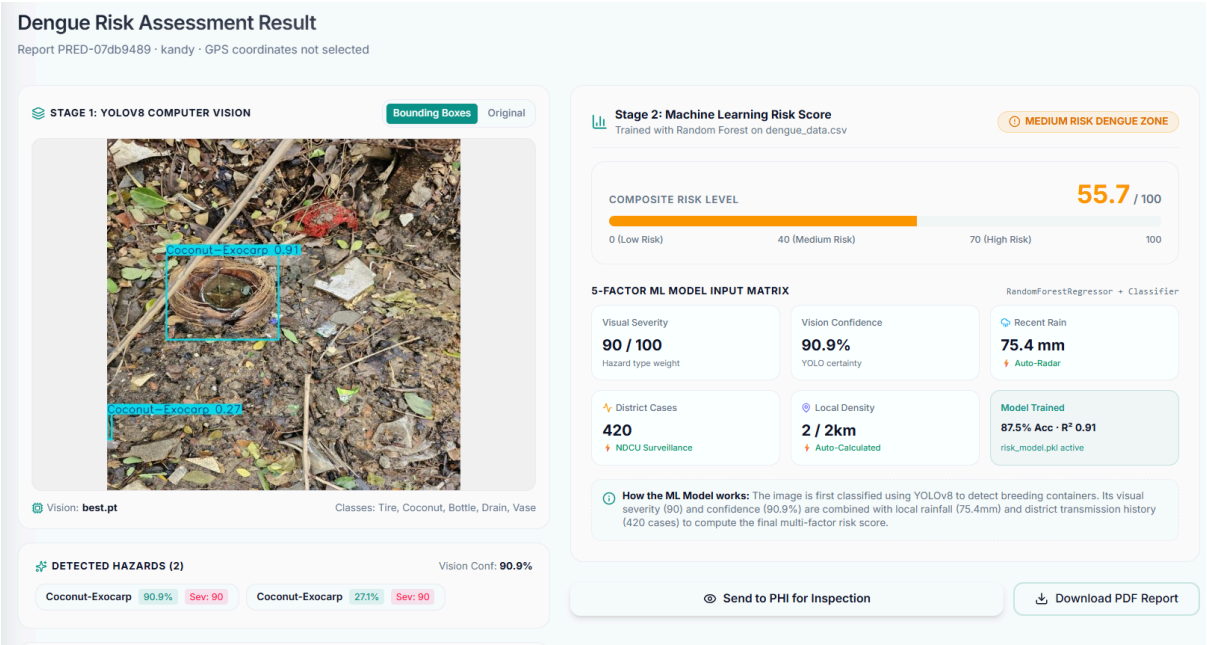


Figure 21: Water-Filled Coconut

11.3 Risk Prioritization Results

The risk-scoring component categorizes reports based on their calculated scores.

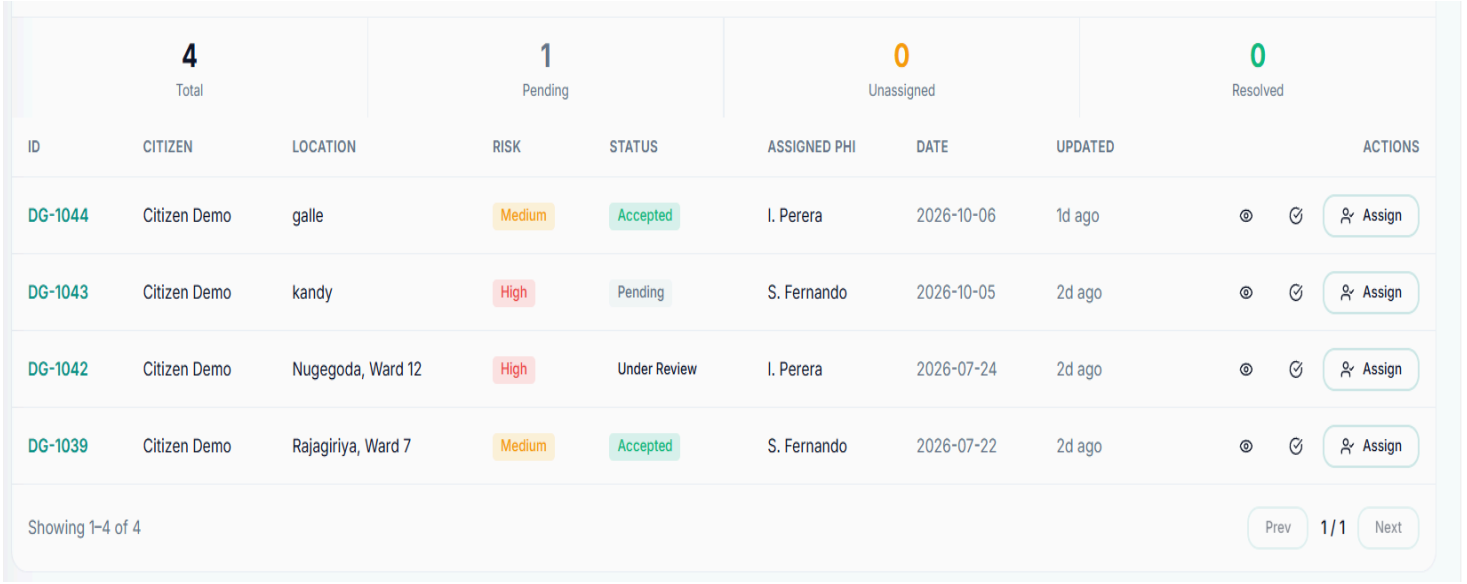


Figure 22: Risk Prioritization in admin panel

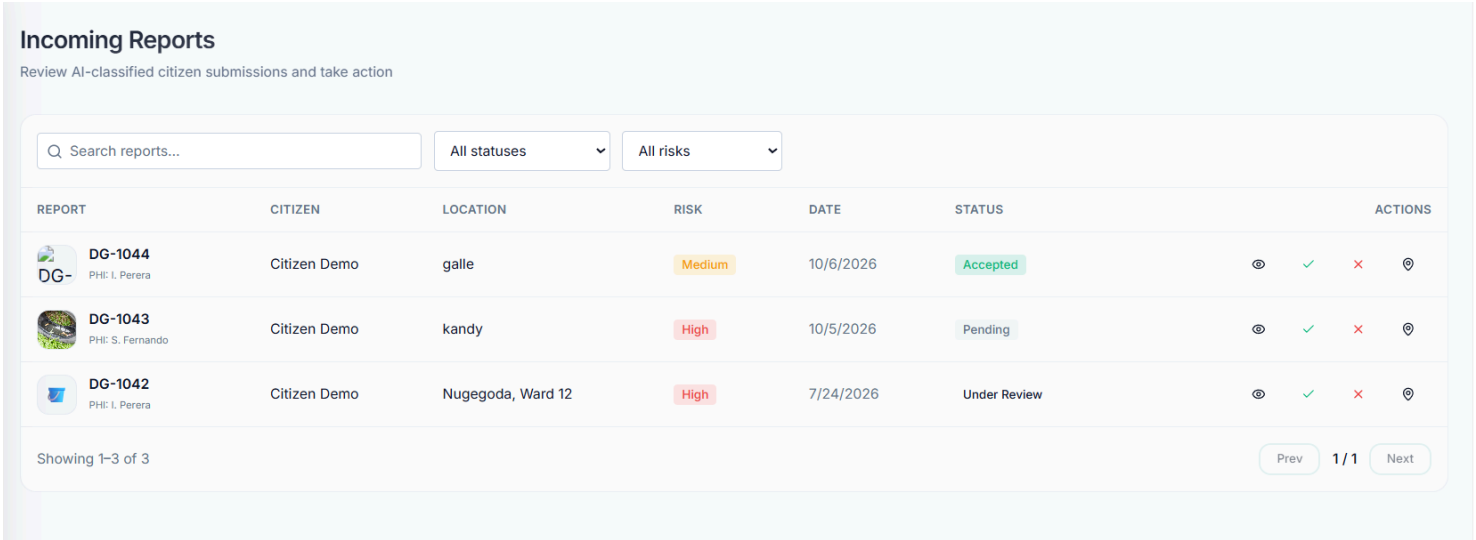


Figure 23: Risk Prioritization in phi panel

12. Model Evaluation

The AI model was evaluated using test images to observe its ability to identify the relevant object classes.

The evaluation should consider:

- Detection accuracy
- Precision
- Recall
- Confidence scores
- False positives
- False negatives

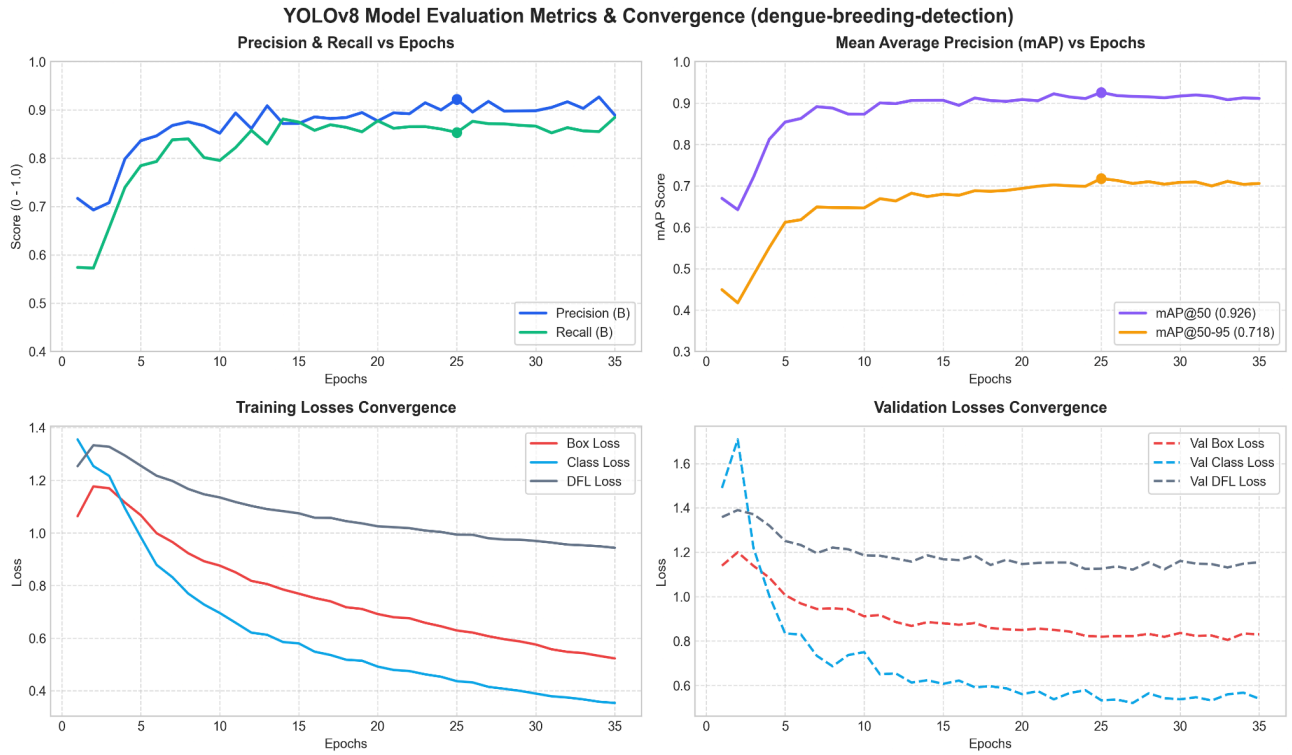


Figure 24: YOLOv8 Model Evaluation Results and Metric Convergence Curves

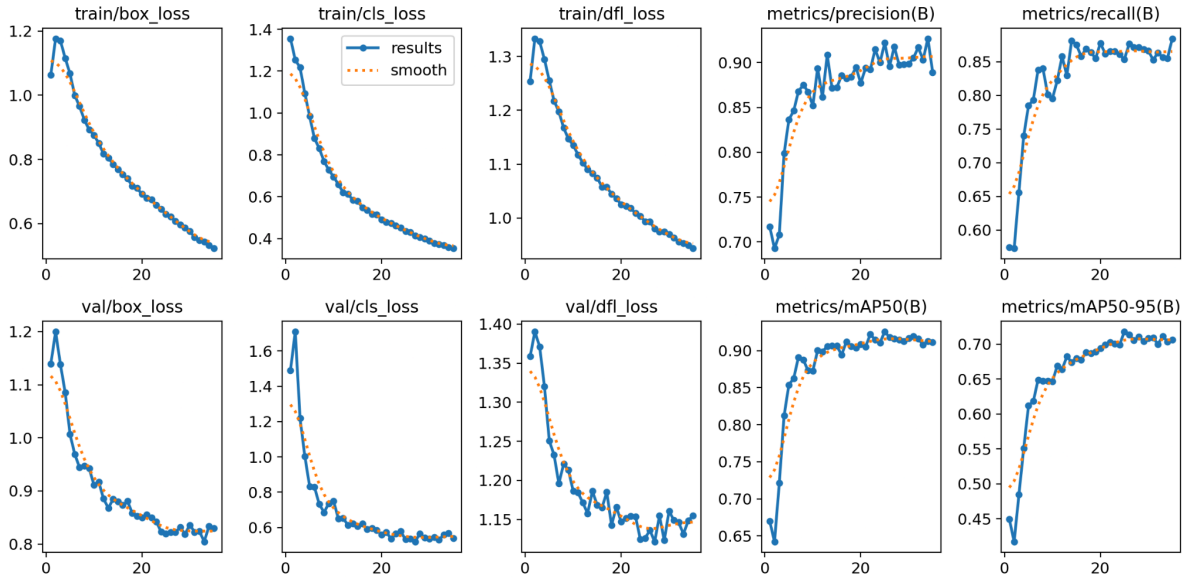


Figure 25: Model Evaluation Results (YOLOv8 Training Summary)

Suggested table:

Metric	Result
Precision	0.9218 (92.2%)
Recall	0.8533 (85.3%)
mAP@50	0.9256 (92.6%)
mAP@50–95	0.7176 (71.8%)

13. Limitations and Discussion

Although the developed system demonstrates the feasibility of AI-assisted breeding-site detection, several limitations remain.

13.1 Dataset Size

The dataset contains 1,250 images. Although this provides a useful starting point, a larger and more diverse dataset could improve model generalization.

13.2 Image Variations

Images can vary significantly depending on:

- Lighting
- Camera angle
- Image quality
- Object size
- Background
- Amount of visible water

These differences can affect detection performance.

13.3 Stagnant Water Detection

No photo alone can tell how long the water has remained there.

It is only the signs of collected or stagnating water that the system finds instead of stating that water has been stagnated for a certain time.

13.4 False Positives and False Negatives

The artificial intelligence may detect a false object or fail to detect an object.

Hence, the artificial intelligence detections cannot be taken to be true breeding sites automatically.

The process of verifying PHI is done to minimize the effects of wrong predictions.

13.5 External Data Availability

The combination of weather, GPS, and dengue data relies upon the presence of high-quality external datasets.
Any discrepancies or missing data from the geographical side will impact risk calculation.

All these shortcomings have been identified at the development phase of this project.

14. Ethical and Social Considerations

The system involves public health information, geographical information, and user-submitted photographs. Therefore, ethical and privacy considerations are important.

14.1 Privacy

User-submitted images and location information should be handled securely. Personal information should not be unnecessarily exposed.

14.2 Data Security

The system should protect stored reports and user information from unauthorized access.

Access to sensitive information should be restricted according to user roles.

14.3 AI Bias

The AI model may perform differently under different environmental conditions, lighting conditions, geographical areas, and image qualities.

A dataset that does not adequately represent real-world environments may lead to biased or less reliable predictions.

14.4 Human Oversight

The system does not replace Public Health Inspectors.

AI predictions are intended to support decision-making. Final verification should be performed by a qualified PHI.

14.5 Responsible Use of AI

The system must not be presented as a medical diagnostic tool.

A detection confidence score must not be interpreted as a probability of dengue infection or a medical risk score.

15. Conclusion

The AI-Based Dengue Breeding Site Detection and Risk Prioritization System illustrates the use of Artificial Intelligence in dengue prevention campaigns.

The system integrates Computer Vision and YOLO object detection to detect the presence of potential mosquito breeding sites from visual indicators. The results are used together with information on location, weather conditions, rainfall, and past information to compute risk scores and priorities.

This system provides a workflow for reporting, where AI analyses pictures uploaded by users, and the risk information is generated for later physical verification by PHIs.

One of the key strengths of the proposed system is the use of AI as a decision support system instead of substituting human decisions. It is necessary due to the fact that just the picture cannot confirm that the water had been stagnant for enough time to become a potential breeding site for mosquitoes.

Thus, the use of AI, Computer Vision, geographic and environmental information, and human verification illustrates how it is possible to create an efficient dengue prevention campaign.

Possible improvements for the project may include increasing the dataset, improving the model's performance, integration of more accurate real-time environmental information, calibrating the risk score and large-scale testing of real reports.

16. References

[1] World Health Organization (WHO). (2025). *Dengue and severe dengue.*
WHO – Dengue and severe dengue

[2] World Health Organization (WHO). (2024). *Dengue and severe dengue – Questions and answers.*
WHO – Dengue Q&A

[3] World Health Organization (WHO). (2016). *Comprehensive Guideline for Prevention and Control of Dengue and Dengue Haemorrhagic Fever.*
WHO – Comprehensive Dengue Guideline

[4] World Health Organization Sri Lanka. (2019). *Preventing Dengue in Sri Lanka.*
WHO Sri Lanka – Preventing Dengue

[5] National Dengue Control Unit, Ministry of Health, Sri Lanka. (2026). *National Dengue Control Unit.*
National Dengue Control Unit – Sri Lanka

[6] National Dengue Control Unit, Ministry of Health, Sri Lanka. (2026). *Disease Surveillance and Risk Assessment.*
NDCU – Disease Surveillance and Risk Assessment

[7] Redmon, J., Divvala, S., Girshick, R., & Farhadi, A. (2016). *You Only Look Once: Unified, Real-Time Object Detection.* Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition (CVPR).
YOLO research paper – arXiv

[8] Ramos, L. T., & Sappa, A. D. (2025). *A Decade of You Only Look Once (YOLO) for Object Detection: A Review.* **IEEE Access, 13**, 192747–192794.
YOLO review paper – arXiv

[9] TensorFlow. (2024). *Convolutional Neural Network (CNN).* TensorFlow Core Documentation.
TensorFlow – CNN Tutorial

[10] Ultralytics. (2026). *Ultralytics YOLO Documentation.*
Ultralytics YOLO Documentation

[11] Scikit-learn. (2026). *Scikit-learn: Machine Learning in Python.*
Scikit-learn Documentation

[12] National Dengue Control Unit, Ministry of Health, Sri Lanka. *Daily Dengue Reports.*
NDCU – Daily Dengue Reports

[13] Sayeedi, M. F. A., & Jisan, M. A. (2025). VisText-Mosquito: A Multimodal Dataset for Mosquito Breeding Site Detection, Surface Segmentation, and Reasoning (Version 3). Mendeley Data. https://doi.org/10.17632/rtsfh7jh7p.3

ESSENTIALS OF ARTIFICIAL INTELLIGENCE

INDIVIDUAL CONTRIBUTION SHEET

Student Name	**W. W. S. N. Wickramarachchi**	**Registration No.**	**D/BIT/24/0043**
Degree / Intake	**Information Technology**	**Group No.**	**58**
Project Title	**AI-Based Dengue Breeding Site Detection and Risk Prioritization System**		

Technical Contributions:

Task / Component	My Contribution	Tools / AI Technique Used	Evidence / Output
Responsive Client Architecture & UI Design System	Built the single-page application structure with dark/light themes, custom CSS variables, and modern responsive layouts.	React 19, Vite, TailwindCSS, CSS Variables	client/src/index.css, App.jsx, responsive layout components
Role-Based Portals (Citizen, PHI, Admin)	Implemented the Citizen complaint wizard, PHI verification interface, and Admin oversight pages with protected routing.	React Router, Context API, Lucide Icons	client/src/pages/ (Citizen, PHI, Admin portal views)
Interactive 3D Mosquito & Prohibition Graphic	Developed an interactive 3D *Aedes aegypti* mosquito model with realistic 80Hz wing flutter, hovering physics, and 360° mouse-drag rotation in front of a 3D red prohibition symbol.	Three.js (WebGL), 3D Procedural Mesh, CatmullRom Curves	client/src/components/3d/Mosquito3D.jsx, home page hero canvas

System Brand Identity & Vector Logo	Designed the custom vector logo featuring a medical blood droplet with an integrated mosquito vector mark.	SVG, JSX, Vector Design	client/src/components/common/BrandLogo.jsx, Navbar & Sidebar logos

Declaration

I confirm that the contributions listed above accurately represent my individual work in this group project, and I can explain or demonstrate the stated work if required.

Student Signature:  **Date: 16/10/2026**

ESSENTIALS OF ARTIFICIAL INTELLIGENCE

INDIVIDUAL CONTRIBUTION SHEET

Student Name	**K.G.T.Nethmina**	**Registration No.**	**D/BSE/25/0025**
Degree / Intake	**Software Engineering/42**	**Group No.**	**58**
Project Title	**AI-Based Dengue Breeding Site Detection and Risk Prioritization System**		

Technical Contributions:

Task / Component	My Contribution	Tools / AI Technique Used	Evidence / Output
Computer Vision & Breeding Site Object Detection	Collected and annotated dataset for 5 high-risk container classes (*Tire, Coconut-Exocarp, Bottle, Drain-Inlet, Vase*). Trained custom YOLOv8 model, tuned confidence & IoU thresholds, and developed OpenCV bounding-box drawing pipeline..	• YOLOv8 (Ultralytics) • Deep Learning / CNN • OpenCV (cv2) • Python	• Model weights file (best.pt) • Annotated images (annotated-{filename}) • Bounding-box spatial coords (xyxy) • Precision, Recall & mAP@50 curves
Multi-Factor Epidemiological Risk Prediction	Designed and trained a dual Random Forest system: **Regressor** (predicts 0–100 risk score) and **Classifier** (*Low/Medium/High* risk). Engineered multi-source feature fusion combining vision severity, confidence, rainfall (mm), NDCU district cases, and report density.	• Scikit-Learn (Random Forest) • Regression & Classification • Multi-Modal Feature Fusion • Pandas & NumPy.	• Serialized model (risk_model.pkl) • Training script (train_rf.py) • Dataset (dengue_data.csv) • R^2 score & classification report
Domain Hazard Severity & Mitigation Engine	Formulated entomological hazard severity weights (Tire: 95, Coconut: 90, Drain: 75, Vase: 70, Bottle: 60). Developed a dynamic rule-based engine mapping detected breeding habitats to	• Knowledge-Based Rules • Heuristic Decision Logic	• Hazard dictionary (CLASS_SEVERITIES) • Dynamic mitigation logic in ai_service.py • Contextual advice output in

	actionable community mitigation recommendations..	• Python Data Structures.	API response
Asynchronous AI Microservice & Zero-Downtime Retraining	Developed an asynchronous REST API microservice on port 5001. Built /predict for real-time inference, /retrain for zero-downtime model updates via dynamic mtime caching, and /health for service telemetry.	• FastAPI & Uvicorn • Joblib (Model Caching) • Asynchronous I/O • REST Architecture	• Backend service (ai_service.py) • Swagger/OpenAPI docs (http://127.0.0.1:5001/docs) • Sub-second response latency logs (< 850ms)
Full-Stack AI Pipeline Integration & Testing	Configured CORS middleware and connected the Python AI microservice with the Express.js application layer and React frontend. Tested end-to-end image ingestion, multipart parsing, and diagnostic result visualization.	• RESTful API Integration • CORS & JSON Schemas • End-to-End Testing	• Route integration (Server/src/routes/aiRoutes.js) • Frontend diagnostic viewer (AIResult.jsx) • Verification screenshots & API test logs

Declaration

I confirm that the contributions listed above accurately represent my individual work in this group project, and I can explain or demonstrate the stated work if required.

Student Signature: _____ **Date:** __10/17/2026_____

ESSENTIALS OF ARTIFICIAL INTELLIGENCE

INDIVIDUAL CONTRIBUTION SHEET

Student Name	**HKDC Akalanka**	**Registration No.**	**D/BIT/24/0074**
Degree / Intake	**41**	**Group No.**	**58**
Project Title	**AI-Based Dengue Breeding Site Detection and Risk Prioritization System**		

Technical Contributions:

Task / Component	My Contribution	Tools / AI Technique Used	Evidence / Output
Server Architecture & REST API	Designed and developed the modular Node.js/Express application server handling citizen reports, PHI operations, and admin requests.	• Node.js • Express.js • REST Architecture	Server/src/app.js, routes/, controllers/
Database Modeling & Media Pipeline	Designed MongoDB schemas (Users, Complaints, Verifications, Notifications) and configured Multer for secure multipart image uploads and storage.	• MongoDB • Mongoose ODM • Multer	Server/src/models/, upload storage directory, database indexes
Authentication & Role-Based Access (RBAC)	Implemented secure token-based authentication with bcrypt password hashing and middleware enforcing permissions across Citizens, PHIs, and Admins.	• JSON Web Tokens (JWT) • Bcrypt • Express Middleware	Server/src/middleware/auth.js, protected API route guards
Intelligent PHI Routing Algorithm	Implemented automated assignment logic that maps incoming citizen	• Geolocation mapping logic • Express Controller	Server/src/controllers/phiController.js, automated assignment logs

	complaints to responsible Public Health Inspectors based on administrative MOH areas and GN divisions.	Logic	

Declaration

I confirm that the contributions listed above accurately represent my individual work in this group project, and I can explain or demonstrate the stated work if required.

Student Signature:  **Date:** ___10/17/2026___

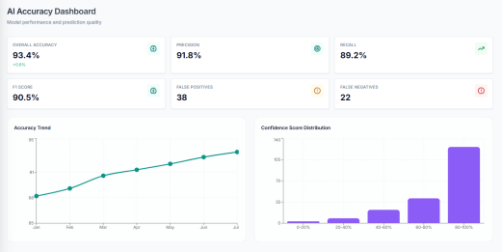
ESSENTIALS OF ARTIFICIAL INTELLIGENCE

INDIVIDUAL CONTRIBUTION SHEET

Student Name	**WMDGJ Warnasooriya**	**Registration No.**	**D/COE/25/0003**
Degree / Intake	**Computer Engineering**	**Group No.**	**58**
Project Title	**AI-Based Dengue Breeding Site Detection and Risk Prioritization System**		

Technical Contributions:

Task / Component	My Contribution	Tools / AI Technique Used	Evidence / Output
Geospatial Mapping and Risk Visualization	Studied how dengue reports and potential breeding sites are displayed on maps using geographical coordinates and colour-coded risk markers.	React, Google Maps , GPS coordinates, location-based risk visualization	GoogleMap.jsx, DengueHotspotMap.jsx, ReportGpsMap.jsx;
Admin Dashboard and Analytics	Studied how dashboard statistics and charts present system information for administrative monitoring.	React, Recharts, data visualization	Dashboard.jsx, Statistics.jsx, TrendLineChart.jsx;
Report Generation and Data Export	Studied how the system generates and exports inspection and report data in different formats.	React, JavaScript, PDFKit, Excel/CSV export utilities	GenerateReports.jsx, MonthlyReports.jsx, exportUtils.js

System Testing and Verification	Reviewed how the frontend, backend, and AI service work together and checked relevant application features.	React, Express.js, Python AI service, functional testing	

Declaration

I confirm that the contributions listed above accurately represent my individual work in this group project, and I can explain or demonstrate the stated work if required.

Student Signature:  **Date:** 17.10.1026